Predicting Ozone Nonattainment in US Counties with Supervised Learning

Machine Learning Mini-Project · UE24CS352A · Section 5E · Team 28
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Abstract

This project investigates supervised machine learning for classifying US counties according to their ozone nonattainment status under the 2015 8-hour ozone National Ambient Air Quality Standard (NAAQS). County-level features were derived from 2025 daily ozone measurements and paired with EPA historical status information. Logistic Regression, Decision Tree, and Random Forest classifiers were evaluated. Random Forest obtained the highest mean F1-score in five-fold cross-validation, while Logistic Regression achieved the highest recall for the nonattainment class.

1. Problem Statement

Ground-level ozone is an important air-quality pollutant. This project examines whether county-level ozone measurement summaries can be used by supervised learning models to classify counties by official ozone nonattainment status. The task is binary classification: class 1 represents a county labelled nonattainment, and class 0 represents a county not labelled nonattainment under the project's target-construction rule.

2. Dataset and Target Construction

Daily ozone observations were obtained from the US Environmental Protection Agency (EPA) AirData dataset for 2025, using ozone parameter code 44201. County status information came from the EPA Green Book historical county-status data. Daily measurements were aggregated by county and joined to status records using state and county Federal Information Processing Standards (FIPS) codes. The working dataset contains 755 county records: 603 class-0 records and 152 class-1 records.

For the 2015 8-hour ozone standard, status values W (whole county) and P (part of county) were treated as positive nonattainment labels. Both indicate nonattainment, differing in geographic extent. Remaining records were assigned class 0 according to the project's label-processing procedure.

Label interpretation. Class 0 means a county was not labelled nonattainment by the implemented processing rule; it does not imply that the county has never had nonattainment status. Maintenance or redesignation status, blank annual fields, and counties absent from the historical listing can affect interpretation. Results therefore depend on the accuracy of label construction.

3. Features and Methodology

Five predictors were used: number of daily ozone records, mean daily ozone concentration, maximum daily ozone concentration, mean daily Air Quality Index (AQI), and mean observation percentage. County identifiers and EPA status fields were excluded from predictors. The target was the binary nonattainment label.

The classifiers were Logistic Regression (standardized inputs and class weighting), Decision Tree (class weighting), and Random Forest (200 trees and class weighting). An initial stratified train/test split was used. Five-fold stratified cross-validation on the training portion compared the models. An additional stratified 80/20 split (random_state=2026) was evaluated for Random Forest.

4. Results

Table 1 reports mean five-fold cross-validation metrics. Precision, recall, and F1-score refer to the positive class (class 1: nonattainment). The final row shows Random Forest on the additional test split.

Table 1. Model performance

Model / evaluation	Accuracy	Precision	Recall	F1-score
Logistic Regression — 5-fold CV	0.781	0.476	0.770	0.588
Decision Tree — 5-fold CV	0.818	0.554	0.533	0.542
Random Forest — 5-fold CV	0.871	0.737	0.566	0.638
Random Forest — additional test split	0.801	0.500	0.533	0.520

4.1 Random Forest: Additional Test Split

Table 2. Confusion matrix for Random Forest

Actual / Predicted	Not nonattainment (0)	Nonattainment (1)
Not nonattainment (0)	105	16
Nonattainment (1)	14	16

On the additional test split, Random Forest achieved 80.1% accuracy, with 0.500 precision, 0.533 recall, and 0.520 F1-score for nonattainment. It correctly identified 16 of the 30 actual nonattainment counties; there were 14 false negatives and 16 false positives.

5. Discussion

In five-fold cross-validation, Random Forest produced the highest mean accuracy (0.871), precision (0.737), and F1-score (0.638). Logistic Regression had the highest recall for the nonattainment class (0.770), identifying a larger share of positive cases at the cost of lower precision (0.476). The additional Random Forest test split yielded an F1-score of 0.520, below its cross-validation mean. This indicates that performance varies across data splits. Because the classes are imbalanced, accuracy alone is insufficient to characterize performance.

6. Conclusion and Limitations

The project implemented a county-level supervised classification workflow using 2025 ozone measurements and EPA ozone-status information. The models identified patterns associated with the constructed labels, but detection of nonattainment counties remained challenging. Random Forest had the highest cross-validation F1-score, while Logistic Regression had the highest cross-validation recall for class 1.

The 2025 measurements and 2025 status labels describe the same year, so this is same-year classification rather than a forecast of future nonattainment. County-level aggregation may conceal variation within counties. Results also depend on the construction and interpretation of historical status labels. The additional test split was created after earlier model and threshold exploration, so it is not a fully untouched final test. Future work should verify target labels, incorporate multiple years of measurements, and evaluate predictions on a later held-out year.

References

[1] US Environmental Protection Agency, “AirData: Air Quality System Data Files,” https://aqs.epa.gov/aqsweb/airdata/download_files.html
[2] US Environmental Protection Agency, “Green Book Data Download,” https://www.epa.gov/green-book/green-book-data-download